

Design and Implementation of Flexible and Reconfigurable SDF-Based FFT Chip Architecture With Changeable-Radix Processing Elements

Xin-Yu Shih[✉], Hong-Ru Chou, and Yue-Qu Liu

Abstract—In this paper, we propose a flexible and reconfigurable changeable-radix fast Fourier transform (FFT) hardware architecture. It aims to support 48 different FFT sizes and up to 4096 FFT points, which are defined in current 3GPP-LTE communication system. The built-up design structure is primarily constructed on a radix-5² basis of single-path delay feedback FFT and up to 18 various changeable radices of FFT processing. A design technique of switchable FIFO usage approach is developed to efficiently manage FIFO arrangement for 48 FFT modes. In addition, a design technique of coarse and fine rotating is designed to effectively reduce twiddle-factor circuit area. By using TSMC 40-nm CMOS technology, an FFT ASIC implementation only has a core area occupation of 0.414 mm² and consumes 49.8 mW in average at maximal working frequency of 526.32 MHz. This innovative design work is competitive as compared to current state-of-the-art works, especially in terms of circuit area cost and power/energy performance evaluation.

Index Terms—Reconfigurable, changeable radix, fast Fourier transform (FFT), single-path delay feedback (SDF), 3GPP-LTE.

I. INTRODUCTION

FAST Fourier transform (FFT) is a most popular design algorithm and hardware module applied for digital communication applications. Initially, in the hardware implementation, the development of processing kernels has many researched outcomes [1]–[4]. Afterwards different FFT system architectures with single FFT-size handling are developed, such as single-path delay feedback (SDF) [5] and [6], multi-path delay feedback (MDF) [7], memory-based processing [8], single-path delay commutator (SDC) [9], multi-path delay commutator (MDC) [10], and so on.

Recently, the communication systems have a necessary request to provide multiple FFT-size supporting, such as 3GPP long-term evolution (LTE) system. There exist several FFT

designs to meet the multi-mode requirement. Reference [11] belongs to MDC approach and uses four parallel processing paths to reduce the computing latency. Reference [12] utilizes a main hybrid SDF/SDC processing path, concatenating SDF- and SDC-based hardware kernels in series. Reference [13] employs memory-based FFT design approach with two primary kernel designs, of which each one can execute radix-2, -3, -4, and -5 FFT processing. As for SDF-based FFT designs in [14]–[21], the calculating datapath mainly consists of concatenated different-radix based FFT operations and each kernel is only responsible for one radix processing, such as radix-2, radix-3, or radix-5. However, [22] can manipulate N FFT points with the mixed powers of 2 and 3 and support 32 FFT modes in 3GPP-LTE applications. Although [23] can handle N FFT points with the mixed powers of 2, 3, and 5, it only offers up to $N = 2048$, which is insufficient for performance requirement of emerging/next-generation applications. For the overall 3GPP-LTE system, it provides many various FFT resolution choices for practical applications, such as the options of short (<100), medium ($100 - 2048$), and long (>2048) FFT lengths. For examples, the design cases with $N > 2048$ (larger FFT analysis resolution) are possibly needed if high-quality performance is really required. Instead, we want to develop a flexible FFT hardware architecture, which achieves as more number of FFT modes as possible (such as 48) and realizes higher FFT points (such as 4096).

The rest of this paper is organized as follows. Section II presents a proposed flexible and reconfigurable changeable-radix FFT hardware architecture. It supports 48 different FFT sizes, which are defined in 3GPP-LTE communication system. Meanwhile, to further elaborate our design, we develop three design techniques for different purposes. a) Changeable-radix processing element is to support 18 radix types of FFT processing by making good use of reconfigurable property. b) Switchable FIFO usage approach is used to arrange very complex usage of 48 various FFT modes, especially for different-radixes based FFT operations in each computing stage. c) Coarse and fine rotating is developed to offer a low-cost hardware solution for supporting large look-up tables (LUTs) of twiddle factor generator in a multi-mode FFT system design. Section III shows ten examples, selected out of 48 FFT sizes, and an ASIC implementation with

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TABLE I
48 FFT MODES IN 3GPP-LTE COMMUNICATION SYSTEM

N	4	8	12	16	24	32	36	48	60	64	72	96	108	120	128 *	144
A	2	3	2	4	3	5	2	4	2	6	3	5	2	3	7	4
B	0	0	1	0	1	0	2	1	1	0	2	1	3	1	0	2
C	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0
Group	E1	E2	E2	E2	E2	E2	E2	E2	E3	E2	E3	E3	E3	E3	E3	E3

N	180	192	216	240	256 *	288	300	324	360	384	432	480	512 *	540	576	600
A	2	6	3	4	8	5	2	2	3	7	4	5	9	2	6	3
B	2	1	3	1	0	2	1	4	2	1	3	1	0	3	2	1
C	1	0	0	1	0	0	2	0	1	0	0	1	0	1	0	2
Group	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3

N	648	720	768	864	900	960	972	1024 *	1080	1152	1200	1296	1536 *	2048 *	3072	4096
A	3	4	8	5	2	6	2	10	3	7	4	4	9	11	10	12
B	4	2	1	3	2	1	5	0	3	2	1	4	1	0	1	0
C	0	1	0	0	2	1	0	0	1	0	2	0	0	0	0	0
Group	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3	E3

Note : (1) FFT Size (N) is represented as $2^A \times 3^B \times 5^C$. (2) * denotes the basic cases in 3GPP-LTE system.

post-automatic place and route (post-APR) results via TSMC 40-nm CMOS technology. Our developed chip compared with other works in current literature is discussed in Section IV. Finally, Section V concludes our proposed chip work with remarkable features.

II. PROPOSED HARDWARE ARCHITECTURE

In this section, we first present a flexible and reconfigurable changeable-radix FFT hardware architecture, supporting up to 4096 FFT points and offering 48 different FFT modes in 3GPP-LTE system. Then, we show three design techniques, including 1) changeable-radix processing element, 2) switchable FIFO usage approach, and 3) coarse and fine rotating.

A. Flexible and Reconfigurable Changeable-Radix FFT Hardware Architecture

TABLE I shows that 3GPP-LTE system defines 48 different FFT sizes (N). Each N is uniquely factorized with 2, 3, and 5, such as $N = 2^A \times 3^B \times 5^C$. For instances, $N = 960$ is represented as $2^6 \times 3 \times 5$, where $(A, B, C) = (6, 1, 1)$ and $N = 1200$ is represented as $2^4 \times 3 \times 5^2$, where $(A, B, C) = (4, 1, 2)$. It is obvious that there are various combinations with the mixed powers of 2, 3, and 5. Therefore, all the design challenges we have to deal with are very huge. Among these FFT sizes, $N = 128/256/512/1024/1536/2048$ show basic cases (downlink usage) whereas other FFT lengths denote extended cases (uplink/downlink usage). Under a bandwidth of 20 MHz, the symbol rate is 18 Msps. By using 64QAM for each symbol, it provides a data rate of 108 Mbps (max). Besides, the sampling rate for $N = 256$ case is 3.84 MHz. In general, for any N FFT points, the sampling rate is expressed as $3.84 \times (N/256)$ MHz.

Fig. 1 is a flexible and reconfigurable changeable-radix FFT hardware architecture. In the main FFT calculating, this system has three design block stages. The system input (SI) is fed into design block stage 1 (DBS 1) and the system output (SO) is at design block stage 3 (DBS 3). Each design block stage has a similar building block structure, including a switchable FIFO (S-FIFO), a FIFO control unit (FCU), a changeable-radix

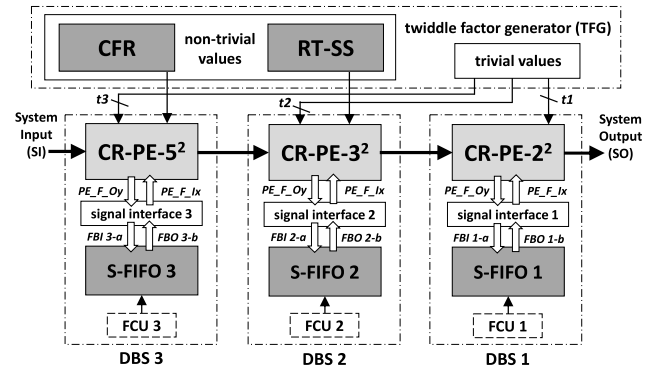


Fig. 1. Proposed FFT hardware architecture.

processing element (CR-PE), and a signal interface between S-FIFO and CR-PE. For different FFT-size processing, we can activate different number of design block stages [see TABLE I again]. E1 indicates that only DBS 1 is activated. E2 means that both DBS 1 and DBS 2 are in use whereas E3 denotes that all three design block stages are at “turn-on” mode.

CR-PE- 5^2 is the reconfigurable processing element, supporting up to radix- 5^2 based FFT operation and 18 different changeable radix types. CR-PE- 3^2 is regarded as the circuit-reduced version of CR-PE- 5^2 . We can design CR-PE- 5^2 and keep the necessary components to derive CR-PE- 3^2 . It supports up to radix- 3^2 based FFT operation and owns 8 changeable radix types. CR-PE- 2^2 is the function-reduced version of CR-PE- 3^2 . The details are introduced in Section II.B.

S-FIFO is to exchange the data signals with CR-PE circuit at the same design block stage. All the FIFO sub-banks are well-arranged and switchable. S-FIFO is well-controlled by FCU, which determines switch true/false conditions and access length of each FIFO sub-bank. The details are in Section II.C.

In DBS 2, the data inputs of CR-PE- 3^2 include PE_F_Ix and the result output from CR-PE- 5^2 . The data outputs of CR-PE- 3^2 are PE_F_Oy and the result output to CR-PE- 2^2 . Signal interface 2 has two directions of signal remapping. First, it remaps PE_F_Oy to FBI 2-a, which pass to S-FIFO 2. Second, it remaps FBI 2-b to PE_F_Ix, which pass to CR-PE- 3^2 . DBS 1 and DBS 3 also have analogous behaviors.

The signal interface status is summarized in TABLE VI and TABLE VII.

At last, twiddle factor generator (TFG) is to generate all the trivial and non-trivial values to the related multipliers in parallel. Trivial multiplication means that one input of the according multiplier has fixed values with simple regularity. It is not taken as actual multiplication. We totally need 3 generator circuits. The whole FFT system has t1, t2, and t3 trivial values for CR-PE-R2², CR-PE-R3², and CR-PE-R5² circuits, respectively. Besides, non-trivial multiplication indicates actual multiplication, which two inputs of the multiplier are non-fixed values. We totally need 2 generator circuits, including the circuits of proposed coarse and fine rotating (CFR) and ROM table with straightforward-storing (RT-SS). Here, we make a focus on developing CFR for producing non-trivial values in DBS 3 [see the details in Section II.D].

B. Changeable-Radix Processing Element

In current literature, there exist several FFT-computing kernel designs, which merge different sizes of butterflies into single reconfigurable butterfly. References [11], [14], and [19] present 2-mode radix-4/8, 2-mode radix-2/3, and 3-mode radix-2/4/8 kernel designs, respectively. Reference [24] develops two kernel designs, such as 2-mode radix-6/8 and 4-mode radix-2/2²/2³/2⁴. Reference [13] shows a reconfigurable 4-mode radix-2, -3, -4, and -5 design, derived from [4]. Even [22] and [23] provides 6 and 7 modes (very limited), respectively. However, the limited operating modes cannot provide further circuit flexibility and any freedom of advanced FFT kernel usages/applications.

To develop a reconfigurable FFT-computing kernel engine with more FFT radix types, we can take a look at TABLE I again. According to various kinds of (A, B, C) combinations, we analyze the relationship of factors (2, 3, and 5) and propose changeable-radix processing element (CR-PE), which is constructed based on a radix-5² FFT basis. The reason why we use radix-5² FFT basis is described as follows. In the core design, we make a good use of trivial multiplication. More trivial multiplications can further reduce area cost. Take radix-5² FFT processing for examples. In the computing aspect, it is a trivial multiplication (without full-range multiplication) between two sub-stages of radix-5 FFT operations. It needs less hardware area because it has less complex multiplication and addition operations with respect to radix-25 FFT processing according to the analysis in [22] and [23]. So, we develop a concatenation of mixed radix-X and radix-Y FFT processing, instead of radix-Z FFT processing (where Z = X × Y).

As listed in TABLE II, CR-PE-5² has 18 radix types. They can be primarily separated into six design categories. In the main hardware resource, CR-PE-5² is composed of 36 complex adders and 10 complex multipliers totally. The main hardware resource of complex adders and multipliers utilized for all 18 radix types is as shown with the star-marked icons of Fig. 2. Different radix types have various adder and multiplier utilization status. For instances, TA-2 fully utilizes all of 36 complex adders and 10 complex multipliers in CR-PE-5², owing both adder and multiplier utilization ratios

TABLE II
DEFINITION OF 18 RADIX TYPES IN CR-PE-5²

Radix Type Notation	Range of k	Radix Type Definition of CR-PE	NTM	Number of Radix Type
TA-k	1 - 2	radix-5 ^k	k-1	2
TB-k	1 - 3	radix-3 ^k	k-1	3
TC-k	1 - 6	radix-2 ^k	k-1	6
TD-k	1 - 2	cat (radix-5, radix-3 ^k)	k	2
TE	(none)	cat (radix-3 ² , radix-2)	2	1
TF-k	1 - 4	cat (radix-3, radix-2 ^k)	k	4

Note 1. cat (radix-x, radix-y) = concatenate radix-x processing with radix-y processing

2. NTM = Number of trivial multiplication

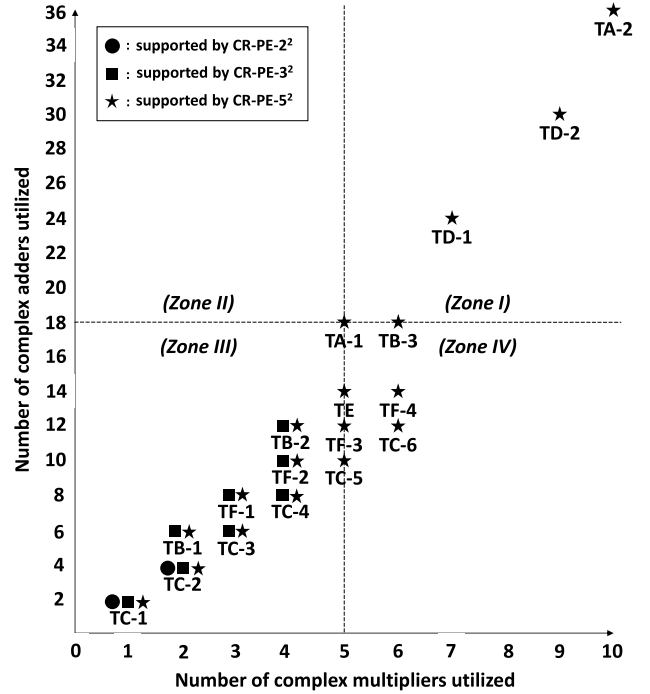


Fig. 2. Adder and multiplier utilization status for different radix types in three kinds of CR-PE circuits.

of 100.0%. TC-6 requires to activate 12 complex adders and 6 complex multipliers for use, delivering adder utilization ratio of 33.3% and multiplier utilization ratio of 60.0%. TB-1 only activates 6 complex adders and 2 complex multipliers on demand. Accordingly, the adder utilization ratio is 16.7% and the multiplier utilization ratio is 20.0%. As a result, both the adder and multiplier utilization ratios are higher than 50% for the radix types in Zone I (TA-2, TD-1, and TD-2). For the radix types in Zone III, both the adder and multiplier utilization ratios are lower than 50%. For the radix types in Zone IV, the multiplier utilization ratio is higher than 50%, but the adder utilization ratio is lower than 50%. Besides, the square-marked icon in Fig. 2 denotes the radix types supported by CR-PE-3². The star-marked icon in Fig. 2 indicates the radix types supported by CR-PE-2². CR-PE-3² and CR-PE-2² have 8 and 2 radix types, respectively. Furthermore, as for

TABLE III
RADIX TYPE SETTING OF THREE CR-PE CIRCUITS FOR 48 FFT MODES

FFT Size	4	8	12	16	24	32	36	48	60	64	72	96	108	120	128	144
CR-PE-2 ²	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2
CR-PE-3 ²	(OFF)	TC-1	TB-1	TC-2	TF-1	TC-3	TB-2	TF-2	TB-1	TC-4	TF-1	TC-3	TB-2	TF-1	TC-4	TF-2
CR-PE-5 ²	(OFF)	(OFF)	(OFF)	(OFF)	(OFF)	(OFF)	(OFF)	(OFF)	TA-1	(OFF)	TB-1	TB-1	TB-1	TA-1	TC-1	TB-1
FFT Size	180	192	216	240	256	288	300	324	360	384	432	480	512	540	576	600
CR-PE-2 ²	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2
CR-PE-3 ²	TB-2	TC-4	TF-1	TF-2	TC-4	TC-3	TB-1	TB-2	TF-1	TC-4	TF-2	TC-3	TC-4	TB-2	TC-4	TF-1
CR-PE-5 ²	TA-1	TB-1	TB-2	TA-1	TC-2	TB-2	TA-2	TB-2	TD-1	TF-1	TB-2	TD-1	TC-3	TD-1	TB-2	TA-2
FFT Size	648	720	768	864	900	960	972	1024	1080	1152	1200	1296	1536	2048	3072	4096
CR-PE-2 ²	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2	TC-2
CR-PE-3 ²	TF-1	TF-2	TC-4	TC-3	TB-2	TC-4	TB-2	TC-4	TF-1	TC-4	TF-2	TF-2	TC-4	TC-4	TC-4	TC-4
CR-PE-5 ²	TB-3	TD-1	TF-2	TB-3	TA-2	TD-1	TB-3	TC-4	TD-2	TE	TA-2	TB-3	TF-3	TC-5	TF-4	TC-6

the radix types of three CR-PE circuits at 48 FFT modes, we can see the detailed setting, as listed in TABLE III.

In addition to the discussion of radix-type supporting and hardware resource, we can zoom-in the detailed hardware circuit of CR-PE-5², as demonstrated in Fig. 3. An entire hardware resource includes 10 complex multipliers, 36 complex adders, 24 right-shifted units, 21 pipelined registers (D flip-flops), and several controlled multiplexers (MUXes). The critical-path timing is dominated by one complex multiplier, four complex adders, and eight MUXes concatenated in series. PE_F_I1 - PE_F_I22 indicate the data signals transferred from FIFO to PE whereas PE_F_O1 - PE_F_O22 denote the data signals transferred from PE to FIFO. Our design totally needs only two non-trivial multiplications with the twiddle factors (W_{nt2} and W_{nt1}), which are located at the last multipliers of CR-PE-5² and CR-PE-3², respectively [see the multiplier marked with dotted-line box in Fig. 3]. All of the right-shifted units with right-shifted amounts (S1 - S20) are arbitrarily determined by the chip users and the possible values are selected in a set of {0, 1, 2, 3}. The purpose is to make the chip users easily adjust signal-to-quantization ratio (SQNR) needed to meet the FFT system requirement. Finally, once finishing CR-PE-5² circuit, we can keep the necessary multiplication and addition circuit operations and remove the redundant circuit components to easily derive CR-PE-3² and CR-PE-2² hardware designs.

C. Switchable FIFO Usage Approach

In the conventional single-mode SDF-FFT nature, FIFO required has a total length of (N-1) for N-point no matter what radix type is used [5] and [6]. Take 4096 FFT points for examples. A total FIFO length is always 4095 no matter radix-2, radix-4, radix-2², radix-2⁴, or other-radix based FFT-processing is adopted. In current literature, there exist seldom discussions of data management for SDF-based or SDF-like FFT designs, only [19] and [25]. A 6-mode FFT system in [19] develops delay-line buffers. Its operation includes not only FFT sizes with power of 2, but also 1536 FFT points. For supporting 1536 FFT points, it modifies the signal flow and needs extra data storage elements, which cause a FIFO hardware overhead ratio of 17/2047. Instead of using FIFO,

[25] presents an in-place memory addressing, but it requires 256 words for 256 FFT points and causes a FIFO hardware overhead of one extra word.

While supporting 48 different FFT sizes, the traffic between the processing element and storage element for each block stage is very complicated. Thus, we develop a smart scheme (switchable FIFO usage approach) to efficiently handle FIFO management. There are two main contributions. 1) We want to keep a total FIFO length of 4095 without any extra FIFO hardware overhead if supporting 4096 FFT points maximally. 2) By detailed analysis for each stage FFT-calculating, the length of each FIFO sub-bank belongs to the power of 4, such as 1, 4, 16, 64, 256, and 1024. Via this systematic way of FIFO partition and arrangement, the processing element in each stage can make the FIFO writing/reading access possible and smooth. In this way, we only adjust the activated FIFO lengths of each FIFO sub-bank to achieve FIFO-bank reusing concept.

Fig. 4 shows that hardware circuits of S-FIFO 1, S-FIFO 2, and S-FIFO 3 are used for data storage of DBS 1, DBS 2, and DBS 3, respectively. FBI denotes the input of FIFO sub-bank whereas FBO represents the output of FIFO sub-bank. Fig. 4(a) depicts that S-FIFO 1 owns three FIFO sub-banks, of which each has a FIFO length of 1. F1-2 is an independent writing/reading access whereas F1-1 and F1-3 are always connected together. All of three FIFO sub-banks are controlled by FCU 1. As shown in Fig. 4(b), there are six FIFO sub-banks located in a 2×3 geometric space, including FIFO lengths of {4, 4, 4, 16, 16, 16}. In these smart FIFO management, we need two switches (SW2-1 and SW2-2) to fulfill the necessary data-exchanged tasks for realizing different FFT sizes. The switch controlling methodology is illustrated in TABLE IV and implemented in FCU 2. All of the controlled conditions are classified into four situations. Take “Situation 2-2” for examples. Meanwhile, the radix type of CR-PE-3² is TC-k, $k = 1 - 4$. All the FIFO sub-banks have FIFO full-access lengths. Both SW2-1 and SW2-2 are asserted (true values). Correspondingly, when F2-3 and F2-1 are connected in series, the input and output ends refer to FBI 2-3 and FBO 2-1, respectively. So do F2-4 and F2-6, having the similar writing/reading behaviors. Besides, we also take an insight into “Situation 2-3”. SW2-1 and SW2-2 are both de-asserted (false values). F2-1, F2-3, and F2-6 are

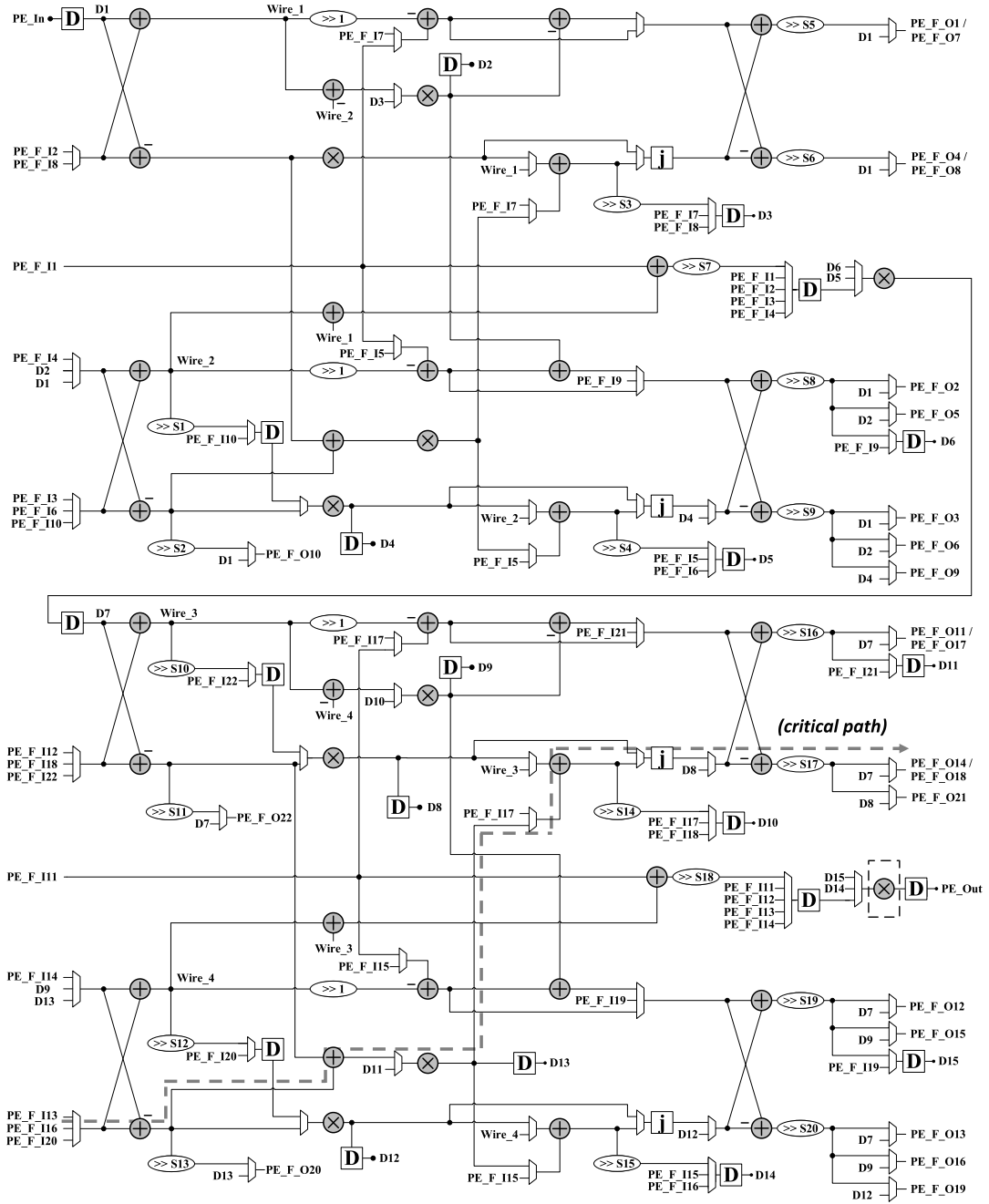


Fig. 3. Hardware circuit of changeable-radix processing element (CR-PE-5²). PE_F_O_y (y = 1-22) signals are passed to signal interface and re-mapped to FBI 3-a (a = 1-9) signals in Fig. 4(c). PE_F_I_x (x = 1-22) signals are passed from signal interface and re-mapped to FBO 3-b (b = 1-9) signals in Fig. 4(c).

TABLE IV
DESIGN OF FCU 2

Situation	Radix-Type of CR-PE-3 ²	Switch Values		FCU 2 : Control Design for S-FIFO 2					
		SW2-1	SW2-2	F2-1	F2-2	F2-3	F2-4	F2-5	F2-6
2-1	TB-k, k=1-2	0	0	(OFF)	4 *	4 *	12	12	(OFF)
2-2	TC-k, k=1-4	1	1	4 *	4 *	4 *	16 *	16 *	16 *
2-3	TF-1	0	0	(OFF)	4 *	(OFF)	8	8	(OFF)
2-4	TF-2	1	0	4 *	4 *	4 *	16 *	16 *	(OFF)

Note : * = writing/reading with FIFO full-access length

turn-off. F2-2 has a FIFO access length of 4 whereas both F2-4 and F2-5 have FIFO access lengths of 8. At the same time, CR-PE-3² circuit is operated at TF-1 radix type.

As for S-FIFO 3 [see Fig. 4(c)], there are nine FIFO sub-banks located in a 3×3 geometric space, including three ones with lengths of 64, three ones with lengths of 256, and three

TABLE V
DESIGN OF FCU 3

Situation	Radix-Type of CR-PE-5 ²	Switch Values			FCU 3 : Control Design for S-FIFO 3									Possible Values of Np
		SW3-1	SW3-2	SW3-3	F3-1	F3-2	F3-3	F3-4	F3-5	F3-6	F3-7	F3-8	F3-9	
3-1	TA-k, k=1-2	0	0	0	Np	Np	Np	Np	(OFF)	5Np	5Np	5Np	5Np	{12, 24, 36, 48}
3-2	TB-k, k=1-3	0	0	0	(OFF)	Np	Np	3Np	3Np	(OFF)	(OFF)	9Np	9Np	{24, 32, 36, 48, 64}
3-3	TC-k, k=1-6	1	1	1	64 *	64 *	64 *	256 *	256 *	256 *	1024 *	1024 *	1024 *	-
3-4	TD-1	0	0	0	(OFF)	Np	Np	(OFF)	(OFF)	3Np	3Np	3Np	3Np	{24, 32, 36, 48, 64}
3-5	TD-2	0	0	0	(OFF)	24	24	72	72	216	216	216	216	-
3-6	TE	0	0	0	(OFF)	64 *	(OFF)	128	128	(OFF)	(OFF)	384	384	-
3-7	TF-1	0	0	0	(OFF)	64 *	(OFF)	128	128	(OFF)	(OFF)	(OFF)	(OFF)	-
3-8	TF-2	1	0	0	64 *	64 *	64 *	256 *	256 *	(OFF)	(OFF)	(OFF)	(OFF)	-
3-9	TF-3	1	0	0	64 *	64 *	64 *	256 *	(OFF)	(OFF)	(OFF)	512	512	-
3-10	TF-4	1	1	0	64 *	64 *	64 *	256 *	256 *	256 *	(OFF)	1024 *	1024 *	-

Note : * = writing/reading with FIFO full-access length

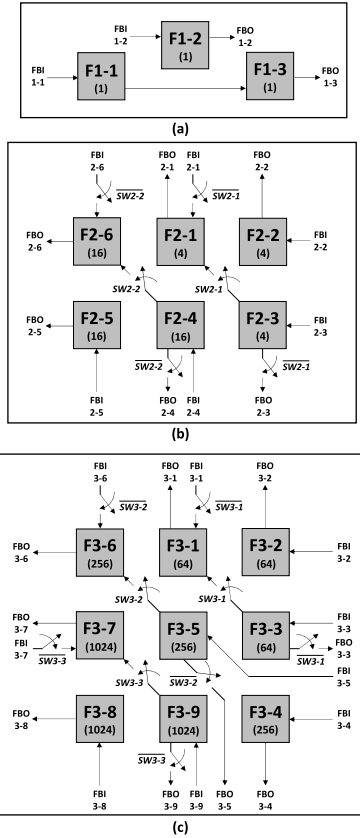


Fig. 4. Hardware circuits of proposed switchable FIFO usage approach: (a) S-FIFO 1, (b) S-FIFO 2, and (c) S-FIFO 3.

ones with lengths of 1024. It needs three switches (SW3-1, SW3-2, and SW3-3). The according FIFO control scheme (FCU 3) is shown in TABLE V. Np denotes the FFT size, supported in previous two design block stages.

After introducing CR-PE and S-FIFO, there is a signal interface between two circuits within every design block stage. As for signal interface 1, it is very simple that PE_F_O19 and PE_F_O20 are connected to FBI 1-2 and FBI 1-1, respectively. FBO 1-2 and FBO 1-3 are connected

to PE_F_I19 and PE_F_I20, respectively. Other signals in CR-PE-2² circuit are unused, just setting as all 0's. For signal interface 2, TABLE VI presents all of connection status, highly depending on different radix types of CR-PE-3² circuit. Take TF-2 radix type for examples. PE_F_O5, PE_F_O6, PE_F_O19, and PE_F_O20 are connected to FBI 2-5, FBI 2-4, FBI 2-2, and FBI 2-3, respectively. FBO 2-1, FBO 2-2, FBO 2-4, and FBO 2-5 are connected to PE_F_I20, PE_F_I19, PE_F_I6, and PE_F_I5, respectively. All of the input/output signals in CR-PE-3² and S-FIFO 2, except specified in TABLE VI, are unused. Besides, TABLE VII illustrates all of connection situations for signal interface 3, which is located between CR-PE-5² and S-FIFO 3.

D. Coarse and Fine Rotating

In current literature, there exist several twiddle-factor generating approaches for FFT applications. Reference [26] shows an area-reduced scheme, which only stores twiddle-factor patterns only located on one-eighth of a unit circle. But it is only applied for FFT sizes with power of 2. Both [16] and [27] develop a multiplier-less method with shift-and-add implementation. It can eliminate the occupied area of multipliers, but its control becomes more and more complicated while dealing with more coefficient choices. However, it has a possibility that the path-computing time is larger than purely one multiplier. In addition, CORDIC [28] and [29] is presented for FFT designs on a discussion of signal flow and hardware component analysis only, without the practical synthesis and APR results. In actual FFT chip hardware implementation, it is difficult to utilize CORDIC due to a crucial reason. It belongs to an iterative calculating, such as 10 clock cycles for a precision of 2^{-10} needed. It leads to a longer operating latency. Therefore, it can use 10 pipelining stages for the circuit implementation while producing one specific twiddle factor for every clock cycle under the requirement of SDF-FFT based architecture. Unfortunately, the number of complex adder utilized is accordingly larger, especially for larger precision required.

TABLE VI
SIGNAL INTERFACE 2: CONNECTIONS BETWEEN CR-PE-3² AND S-FIFO 2

Radix-Type of CR-PE-3 ²	"x" Value of PE_F_Ox : Connections from CR-PE-3 ² to S-FIFO 2						"y" Value of PE_F_Iy: Connections from S-FIFO 2 to CR-PE-3 ²					
	FBI 2-1	FBI 2-2	FBI 2-3	FBI 2-4	FBI 2-5	FBI 2-6	FBO 2-1	FBO 2-2	FBO 2-3	FBO 2-4	FBO 2-5	FBO 2-6
TB-1	(U)	15	16	(U)	(U)	(U)	(U)	15	16	(U)	(U)	(U)
TB-2	(U)	15	16	18	17	(U)	(U)	15	16	18	17	(U)
TC-1	(U)	19	(U)	(U)	(U)	(U)	(U)	19	(U)	(U)	(U)	(U)
TC-2	(U)	19	20	(U)	(U)	(U)	20	19	(U)	(U)	(U)	(U)
TC-3	(U)	19	20	(U)	21	(U)	20	19	(U)	(U)	21	(U)
TC-4	(U)	19	20	22	21	(U)	20	19	(U)	(U)	21	22
TF-1	(U)	19	(U)	6	5	(U)	(U)	19	(U)	6	5	(U)
TF-2	(U)	19	20	6	5	(U)	20	19	(U)	6	5	(U)

Note : U denotes "unused" input or output ports.

TABLE VII
SIGNAL INTERFACE 3: CONNECTIONS BETWEEN CR-PE-5² AND S-FIFO 3

Radix-Type of CR-PE-5 ²	"x" Value of PE_F_Ox : Connections from CR-PE-5 ² to S-FIFO 3									"y" Value of PE_F_Iy: Connections from S-FIFO 3 to CR-PE-5 ²								
	FBI3-1	FBI3-2	FBI3-3	FBI3-4	FBI3-5	FBI3-6	FBI3-7	FBI3-8	FBI3-9	FBO 3-1	FBO 3-2	FBO 3-3	FBO 3-4	FBO 3-5	FBO 3-6	FBO 3-7	FBO 3-8	FBO 3-9
TA-1	13	11	12	14	(U)	(U)	(U)	(U)	(U)	13	11	12	14	(U)	(U)	(U)	(U)	(U)
TA-2	13	11	12	14	(U)	1	4	2	3	13	11	12	14	(U)	1	4	2	3
TB-1	(U)	15	16	(U)	(U)	(U)	(U)	(U)	(U)	(U)	15	16	(U)	(U)	(U)	(U)	(U)	(U)
TB-2	(U)	15	16	17	18	(U)	(U)	(U)	(U)	(U)	15	16	17	18	(U)	(U)	(U)	(U)
TB-3	(U)	15	16	17	18	(U)	(U)	5	6	(U)	15	16	17	18	(U)	(U)	5	6
TC-1	(U)	19	(U)	(U)	(U)	(U)	(U)	(U)	(U)	(U)	19	(U)	(U)	(U)	(U)	(U)	(U)	(U)
TC-2	(U)	19	20	(U)	(U)	(U)	(U)	(U)	(U)	20	19	(U)	(U)	(U)	(U)	(U)	(U)	(U)
TC-3	(U)	19	20	21	(U)	(U)	(U)	(U)	(U)	20	19	(U)	21	(U)	(U)	(U)	(U)	(U)
TC-4	(U)	19	20	21	22	(U)	(U)	(U)	(U)	20	19	(U)	21	(U)	22	(U)	(U)	(U)
TC-5	(U)	19	20	21	22	(U)	(U)	9	(U)	20	19	(U)	21	(U)	22	(U)	9	(U)
TC-6	(U)	19	20	21	22	(U)	(U)	9	10	20	19	(U)	21	(U)	22	10	9	(U)
TD-1	(U)	15	16	(U)	(U)	1	4	2	3	(U)	15	16	(U)	(U)	1	4	2	3
TD-2	(U)	15	16	17	18	1	4	2	3	(U)	15	16	17	18	1	4	2	3
TE	(U)	19	(U)	5	6	(U)	(U)	7	8	(U)	19	(U)	5	6	(U)	(U)	7	8
TF-1	(U)	19	(U)	5	6	(U)	(U)	(U)	(U)	(U)	19	(U)	5	6	(U)	(U)	(U)	(U)
TF-2	(U)	19	20	5	6	(U)	(U)	(U)	(U)	20	19	(U)	5	6	(U)	(U)	(U)	(U)
TF-3	(U)	19	20	21	(U)	(U)	(U)	5	6	20	19	(U)	21	(U)	(U)	(U)	5	6
TF-4	(U)	19	20	21	22	(U)	(U)	5	6	20	19	(U)	21	(U)	22	(U)	5	6

Note : U denotes "unused" input or output ports.

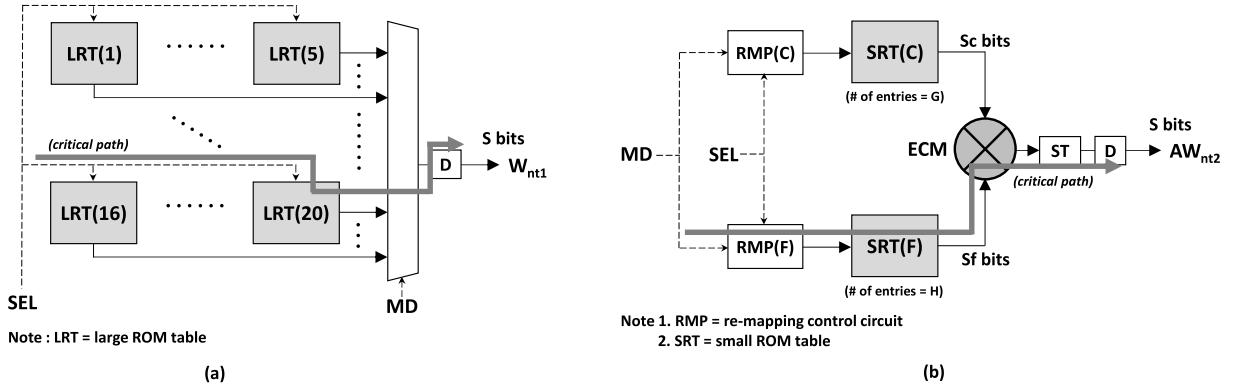


Fig. 5. Twiddle factor generator: (a) ROM table with straightforward-storing (RT-SS) and (b) coarse and fine rotating (CFR).

As mentioned earlier [see Section II.B], our design requires two non-trivial multiplications with the twiddle factors, W_{nt1} and W_{nt2} . Fig. 5(a) depicts a direct twiddle-factor generating

approach, ROM table with straightforward-storing (RT-SS). LRT refers to a large ROM table, which directly stores plenty of twiddle-factor values. Each twiddle factor consists of

TABLE VIII
ROM TABLE ENTRY ANALYSIS OF RT-SS ON W_{nt2}

Note : NRTE = Number of ROM table entry

LRT(i)	FFT Size (N)	NRTE Utilized	LRT(i)	FFT Size (N)	NRTE Utilized
1	4096/2048/1024/512/256/128	1238	11	864/288/96	325
2	3072	1108	12	768	352
3	1536	639	13	720	321
4	1296/432/144	522	14	648/216/72	234
5	1200/240	485	15	600/120	209
6	1152/384	505	16	576/192	269
7	1080	437	17	540	230
8	972/324/108	367	18	480	208
9	960	425	19	360	156
10	900/180	347	20	300/60	132
Total NRTE =			8509		

TABLE IX
ERROR ANALYSIS AND SQNR DEGRADED DUE TO DESIGN TECHNIQUE OF CFR

LRT(i)	Max Angular Error (10^{-3} rad)	Max Rotation Error (10^{-3})	System SQNR Degraded (dB)	LRT(i)	Max Angular Error (10^{-3} rad)	Max Rotation Error (10^{-3})	System SQNR Degraded (dB)
1	0.97	1.32	0.48	11	1.59	1.80	0.35
2	1.54	1.65	0.20	12	1.54	1.65	0.79
3	1.54	1.65	0.59	13	1.62	1.80	0.20
4	1.66	1.94	0.19	14	1.53	1.72	0.71
5	1.60	1.87	0.36	15	1.60	1.80	0.36
6	1.60	2.10	0.02	16	1.60	1.89	0.15
7	1.53	1.72	0.30	17	1.52	1.63	0.43
8	1.52	1.80	0.62	18	1.54	1.79	0.25
9	1.54	1.79	0.44	19	1.53	1.72	0.85
10	1.56	1.71	0.36	20	1.56	1.55	0.43
AVG.					1.53	1.75	0.40

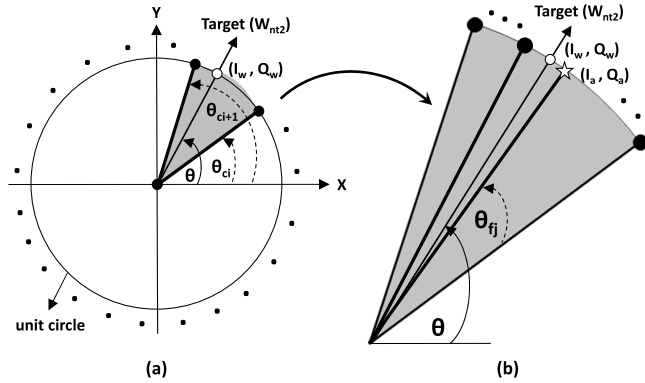


Fig. 6. Design concept of coarse and fine rotating (CFR). (a) Coarse rotating. (b) Fine rotating.

S-bit real and imaginary parts. While supporting more FFT sizes, the number of LRTs needed gets increasing. Thus, this approach can be only applied for producing W_{nt1} because it needs less number of ROM table entries. But it is not suitable for producing W_{nt2} because it requires extremely large number of ROM table entries. From TABLE VIII, there are 39 FFT modes needed for producing W_{nt2} . For instances, LRT(4) is a shared large ROM table, which is used for $N = 1296, 432$, and 144. It actually stores 522 table entries, instead of 1296. In other words, by hardware sharing, we still have to build up twenty ROM tables, LRT(1) - LRT(20), to produce the final output (W_{nt2}) by SEL (counter select) and MD (mode). The total number of ROM table entries is pretty large, such as 8509. As a result, it obviously not only occupies large hardware area, but also causes maintenance difficulty. Thus, we develop a design technique of coarse and fine rotating (CFR), which owns much smaller area cost under acceptable error extents.

As drawn in Fig. 6, we first assume that the target twiddle factor value (W_{nt2}) has an angle, θ , with respect to positive x-axis. Then, the main design concept is specified as follows.

1) *First Level (Coarse Rotating)*: Fig. 6(a) shows that W_{nt2} is located in a certain position on a unit circle with x-y coordinates of (I_w, Q_w). W_{nt2} is marked with a hollow-circle icon. We can equally divide this unit circle into G zones with

uniform boundaries, connected from the origin. The uniform boundaries with respect to positive x-axis have individual angles, denoted as $\theta_{c0} - \theta_{c(G-1)}$. We can determine the coarse angle (θ_{ci}), which exactly satisfies the inequality of (1).

$$\theta_{ci} \leq \theta \leq \theta_{ci+1}, \quad \text{where } i \in \{0, 1, \dots, G-1\} \quad (1)$$

2) *Second Level (Fine Rotating)*: Once the coarse angle (θ_{ci}) is selected/determined, we can pick out the circular sector area, as shaded in gray in Fig. 6(a). Again, we can equally divide this gray-colored area into H sub-zones, as shown in Fig. 6(b). As the angle definition in first-level rotating, we also have H angles, $\theta_{f0} - \theta_{f(H-1)}$, in an analogous manner. The fine angle (θ_{fj}) is suitably determined while matching the condition of (2).

$$j = \arg(\min|\theta - \theta_{fm}|), \quad \text{where } m \in \{0, 1, \dots, H-1\} \quad (2)$$

Therefore, the task of searching for the approximate angle is totally completed, such as $\theta \approx \theta_{ci} + \theta_{fj}$. Correspondingly, the x-y coordinates of this approximate angle are (I_a, Q_a), marked with a hollow-star icon in Fig. 6(b).

In the following fixed-point discussion and performance analysis, we choose $G = H = 64$. W_{nt2} has a wordlength of 14 bits for both real and imaginary parts. Rotating in both first and second levels behaves 12-bit I/Q representation. In order to make a comparison, we can utilize the existing performance indexes/indicators to evaluate, such as angular error [27], rotation error [30], and signal-to-quantization-noise ratio (SQNR) [31]. We first assume that the accurate twiddle-factor point is P , the approximate point is P' and the origin is O . θ is the angle between $\vec{OP}$ and positive X-axis whereas θ' is the angle between $\vec{OP'}$ and positive X-axis. Angular error means the error in the angle point-of-view, referring to the angle difference between θ and θ' . If angular error gets smaller, it indicates the approximate point (P') is closer to the accurate twiddle-factor point (P). In addition, rotation error means the error in the distance point-of-view, referring to the distance between P and P' in a 2-dimensional XY plane. It is similar that P' is closer to P while rotation error gets smaller. Therefore, these two performance indexes/indicators are to evaluate the twiddle-factor error in different perspectives.

TABLE IX lists all the performance approximation of the original large tables, LRT(1) - LRT(20). LRT(4) is originally

TABLE X
ERROR ANALYSIS AND SYNTHESIS AREA RESULTS OF DIFFERENT BITS FOR CFR DESIGN TECHNIQUE APPLIED FOR W_{nt2}

(S, Sc, Sf)		(12, 12, 12)		(14, 12, 12)		(16, 13, 13)		(18, 14, 14)	
RT-SS		47130.4		50989.7		54848.9		58708.2	
CFR	RMP(C)	3030.8 (35.9%)	8437.7	3030.8 (35.8%)	8460.4	3030.8 (33.6%)	9013.0	3030.8 (32.5%)	9319.7
	RMP(F)	2479.8 (29.4%)		2479.8 (29.3%)		2479.8 (27.5%)		2479.8 (26.6%)	
	SRT(C)	344.3 (4.1%)		344.3 (4.1%)		363.6 (4.0%)		383.1 (4.1%)	
	SRT(F)	262.9 (3.1%)		262.9 (3.1%)		293.8 (3.3%)		315.7 (3.4%)	
	ECM	2319.9 (27.5%)		2342.6 (27.7%)		2845.0 (31.6%)		3110.3 (33.4%)	
Area Saving		38692.7		42529.3		45835.9		49388.5	
Area Saving Ratio		82.10%		83.41%		83.57%		84.13%	
Average Rotation Error (10 ⁻⁴)		6.30		6.15		4.43		3.86	

Note : area unit = μm^2

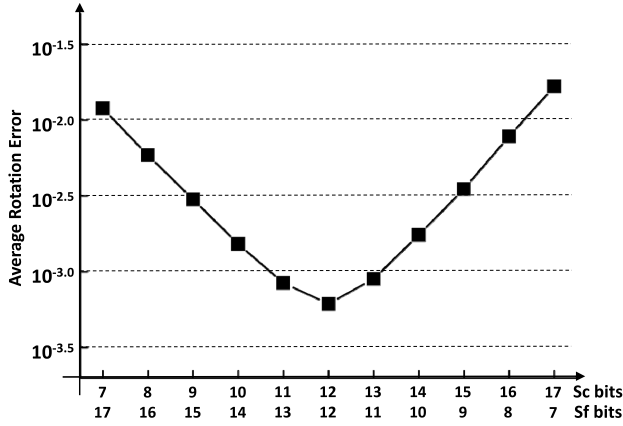


Fig. 7. Average rotation error analysis of different (Sc, Sf)-bit combinations.

the shared ROM table for $N = 1296, 432$, and 144 . LRT(6) is originally the shared ROM table for $N = 1152$ and 384 . Here, max angular error indicates the worst case on angular error and max rotation error denotes the worst case on rotation error. Among all of the original large tables, LRT(4) has the max angular error (1.66×10^{-3} in rad) and LRT(6) has the max rotation error (2.10×10^{-3}). In addition, for the whole FFT system performance, we can evaluate system output (SO) in terms of SQNR degraded in dB. LRT(19) is only used for $N = 360$ and has the largest SQNR degradation with respect to RT-SS approach, such as 0.85 dB. The SQNR degradation in average is 0.40 dB. The performance degradation at system output (SO) due to CFR technique can be negligible in reality.

Furthermore, we assume that coarse and fine rotating have Sc and Sf bits, respectively, in the fixed-point representation. We can vary different Sc-bit and Sf-bit combinations for evaluating the system performance, as drawn in Fig. 7. Here, the rotation error indicates the error between (I_w, Q_w) and (I_a, Q_a) in x-y plane. Under the assumption of keeping $St = Sc + Sf (= 24 \text{ bits})$ the same, the simulation case of $Sc = Sf = St/2 (= 12 \text{ bits})$ has a best performance in terms of average rotation error. If the absolute difference between Sc and Sf becomes large, the average rotation error increases. While choosing other values of St, it also has similar phenomenon. Consequently, we decide to choose the bit setting of $Sc = Sf$ for our chip hardware implementation.

Fig. 5(b) depicts the built-up hardware circuit. There are two small ROM tables (SRT(C) and SRT(F)), which store the coarse and fine rotated angles, respectively. SRT(C) and SRT(F) have entry counts of G and H, respectively. The real or imaginary part of each entry in SRT(C) is of Sc bits. The real or imaginary part of each entry in SRT(F) is of Sf bits. RMP(C), the remapping control circuit for coarse rotating, is responsible for selecting the according target entry by MD and SEL. So does RMP(F) for fine rotating. Extra complex multiplier (ECM) is used to combine two-level rotating. Finally, the approximated output (AW_{nt2}) is directly saturated and truncated (ST) with S-bit wordlength representation. The path-computing time consists of one complex multiplier, one real adder, and seven MUXes concatenated in series.

TABLE X shows error analysis and synthesis area results with TSMC 40-nm CMOS technology. If $G = 64$ and $H = 64$, the number of total ROM table entries is reduced from 8509 to 128 ($= 64 + 64$), owning a large entry reduced ratio of 98.5%. As for CFR with $S = 14$ and $Sc = Sf = 12$, sub-circuits include SRT(C), SRT(F), RMP(C), RMP(F), and ECM, occupying 4.1%, 3.1%, 35.8%, 29.3%, and 27.7%, respectively. The area of proposed CFR circuit is only $8460.4 \mu m^2$. To make a fair comparison (design reference), we also use the same process technology and implement the same circuit function by RT-SS approach with $S = 14$. The area occupies $50989.7 \mu m^2$. It is obvious that it can save circuit area by 83.41% (from 50989.7 to $8460.4 \mu m^2$) while the average rotation error between AW_{nt2} and W_{nt2} is 6.15×10^{-4} . In the different-bit combinations, the area saving ratio varies from 82.10% to 84.13% under the acceptable rotation error extent of 10^{-4} . As S-bit increases, the area-saving phenomenon is more obvious due to large table entry reduction.

III. DESIGN EXAMPLES AND ASIC IMPLEMENTATION

TABLE XI shows ten examples, which are selected out of 48 FFT sizes. For $N = 648$ case, we suggest the right-shifted amounts, as listed in 8th column. In CR-PE-5², {S4, S8, S9, S14-S17, S19, S20} are chosen as {2, 2, 2, 1, 1, 1, 1, 1}, respectively. In CR-PE-3², {S4, S8, S9, S19, S20} are decided as all 1's whereas setting $S12 = S13 = 1$ is for CR-PE-2². All the right-shifted values, except mentioned here, are 0's. Therefore, at system output (SO),

TABLE XI
10 SELECTED FFT MODES FOR SHOWING SQNR IMPROVEMENT WITH SUGGESTED RIGHT-SHIFTED AMOUNTS

FFT Size (N)		4096		3072		1296		1152		1080		960		768		648		300		48	
Suggested Right-Shifted Amounts	CR-PE-5 ²	S1	2	S4	2	S4	2	S3	2	S5	3	S5	3	S4	2	S4	2	S5	3	(OFF)	
		S2	2	S8	2	S8	2	S4	1	S6	3	S6	3	S8	2	S8	2	S6	3		
		S8	1	S9	1	S9	2	S5	2	S7	3	S7	3	S9	2	S9	2	S7	3		
		S9	1	S10	1	S14	1	S6	2	S8	3	S8	3	S12	1	S14	1	S8	3		
		S10	1	S11	1	S15	1	S8	1	S9	3	S9	3	S13	1	S15	1	S9	3		
		S11	1	S12	1	S16	1	S9	1	S14	1	S15	1	S19	1	S16	1	S16	2		
		S12	1	S13	1	S17	1	S19	1	S15	1	S19	1	S20	1	S17	1	S17	2		
		S13	1	S16	1	S19	1	S20	1	S16	1	S20	1			S19	1	S18	2		
		S16	1	S17	1	S20	1			S17	1					S20	1	S19	2		
		S17	1	S19	1									S19	1						
		S19	1	S20	1									S20	1						
		S20	1																		
	CR-PE-3 ²	S10	1	S10	1	S4	1	S10	1	S4	1	S10	1	S10	1	S4	1	S15	1	S4	2
		S11	1	S11	1	S8	1	S11	1	S8	1	S11	1	S11	1	S8	1	S19	1	S8	2
		S12	1	S12	1	S9	1	S12	1	S9	1	S12	1	S12	1	S9	1	S20	1	S9	2
		S13	1	S13	1	S12	1	S13	1	S19	1	S13	1	S13	1	S19	1			S12	1
		S16	1	S16	1	S13	1	S16	1	S20	1	S16	1	S16	1	S20	1			S13	1
		S17	1	S17	1	S19	1	S17	1			S17	1	S17	1			S19	1		
		S19	1	S19	1	S20	1	S19	1			S19	1	S19	1			S20	1	S20	1
		S20	1	S20	1			S20	1			S20	1	S20	1						
CR-PE-2 ²		S12	1	S12	1	S12	1	S12	1	S12	1	S12	1	S12	1	S12	1	S12	1	S12	1
	S13	1	S13	1	S13	1	S13	1	S13	1	S13	1	S13	1	S13	1	S13	1	S13	1	
SQNR with Shift (dB)		34.81		35.73		49.49		43.49		48.92		42.83		41.73		52.12		50.29		55.57	
SQNR without Shift (dB)		0.11		0.12		0.20		0.21		0.22		0.24		0.26		0.29		0.46		1.47	
SQNR Improved (dB)		34.70		35.61		49.29		43.28		48.70		42.59		41.47		51.83		49.83		54.10	

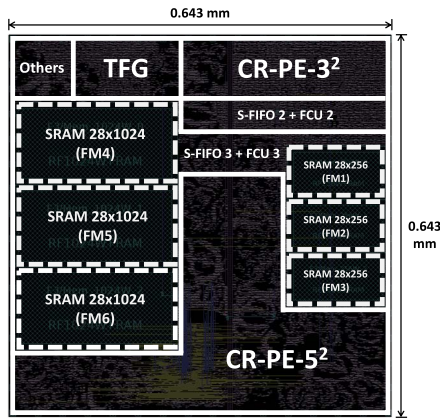


Fig. 8. Chip layout plot of the FFT ASIC implementation.

SQNR with suggested right-shifted values is enhanced from 0.29 to 52.12 dB, i.e., SQNR improved of 51.83 dB. Besides, the other FFT cases with suggestions are treated as references, improving SQNR by 34.7 - 54.1 dB.

Our proposed flexible and reconfigurable 48-mode FFT hardware architecture is first designed and synthesized with Design Compiler tool via TSMC 40-nm CMOS technology. And then, it is implemented with IC Compiler tool. The system input and output both have 14-bit wordlength while all the trivial and non-trivial twiddle factors (both CFR and RT-SS) are also in a 14-bit fixed-point format representation. As the chip layout plot depicted in Fig. 8, the total core area of implemented result is only 0.643 mm × 0.643 mm (= 0.414 mm²), of which the ratios of combinational logics

TABLE XII
APR AREA RESULTS OF PROPOSED FFT HARDWARE ARCHITECTURE

Sub-Circuits of Proposed FFT System			Core Area with TSMC 40-nm Process (um ²)				
			Comb.	Seq.	Total	Occupied Ratio	
PE	CR-PE-2 ²		1,980	624	2,604	0.6%	40.5%
	CR-PE-3 ²		35,405	1,664	37,069	9.0%	
	CR-PE-5 ²		123,785	4,160	127,945	30.9%	
FIFO	S-FIFO 1 + FCU 1		9	624	633	0.2%	51.8%
	S-FIFO 2 + FCU 2		5,934	12,500	18,434	4.5%	
	S-FIFO 3 + FCU 3		22,825	172,482	195,307	47.1%	
TFG	Trivial Values		3,245	2,057	5,302	1.3%	5.1%
	Non-Trivial Values	CFR	12,593	1,259	13,852	3.3%	
		RT-SS	1,619	253	1,872	0.5%	
System Control Unit (SCU)			6,271	4,341	10,612	2.6%	
Total Core Area			413,630			100.0%	

and sequential circuits are 51.7% and 48.3%, respectively. The average power dissipation is 49.8 mW for all 48 FFT modes while the working frequency operates 526.32 MHz maximally.

In the proposed architecture, there totally are eighteen FIFO sub-banks in S-FIFO 1, S-FIFO 2, and S-FIFO 3 [see Fig. 4]. In the actual hardware implementation, there are six SRAM memories, which are used for six FIFO sub-banks in S-FIFO 3. The other twelve FIFO sub-banks are directly implemented with D flip-flops. The SRAM memory sizes include three for 28 × 256 (FM1 - FM3) and three for 28 × 1024 (FM4 - FM6), summing up to 107.52K bits. Each SRAM 28×256 occupies 2.66% of total core area whereas each SRAM 28 × 1024 occupies 7.68% of total core area. In addition, TABLE XII analyzes the implemented area of each sub-circuit.

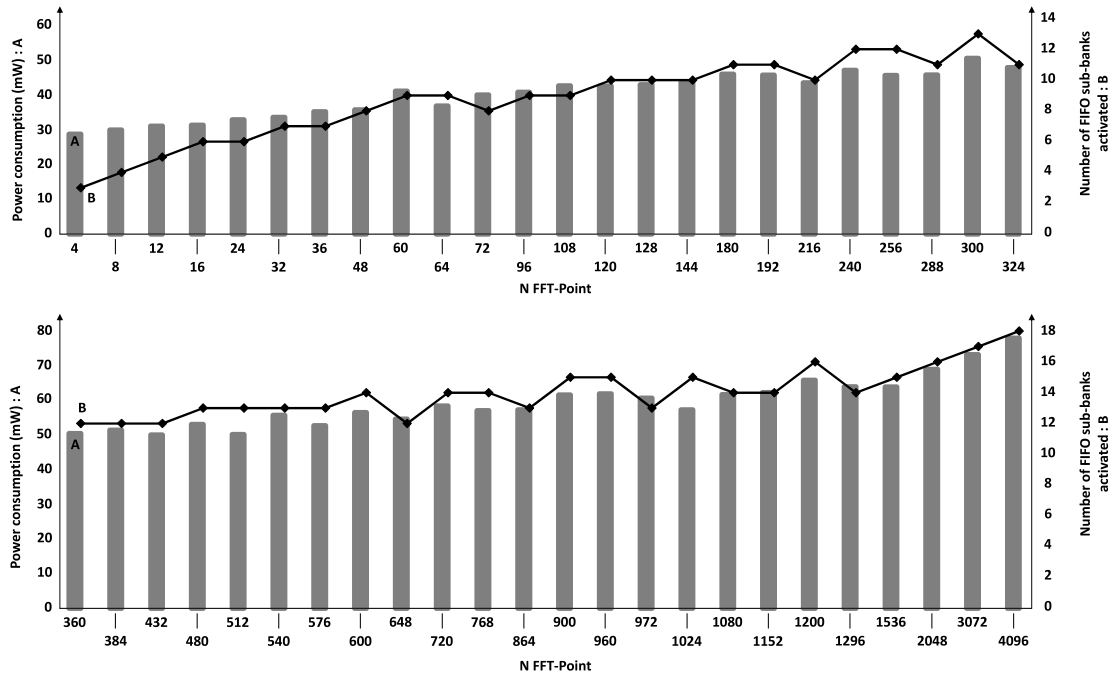


Fig. 9. Power consumption and number of FIFO sub-banks activated.

All of three CR-PE circuits have a core area of 0.168 mm^2 , occupying 40.5% of total core area. Three S-FIFOs with the related control circuits (FCUs) have 51.8% of total core area, such as 0.214 mm^2 . TFG only has a core area occupation of 5.1%. Moreover, Fig. 9 demonstrates the relationship of the power dissipation at maximal working frequency (such as 526.32 MHz) and the individually activated status of FIFO sub-banks needed for all of 48 FFT modes.

IV. CHIP COMPARISON

TABLE XIII lists several multi-mode FFT design works for 3GPP-LTE applications in current literature. In the system level, [11], [12], [19], and [24] only support 4 - 6 basic cases (downlink usages). They cannot provide the extended cases (uplink/downlink usage) of the FFT processing in 3GPP-LTE applications. Reference [22] presents a LEGO-like constructing approach, which can handle the FFT points with the mixed powers of 2 and 3 only. Because the reconfigurable processing elements are only based on radix- 2^3 and radix- 3^2 SDF-FFT basis, it just covers 32 FFT modes defined in 3GPP-LTE applications. Reference [13] is a memory-based FFT architecture, which covers 40 FFT modes and supports up to 2048 FFT points. It shows two individual circuit cores, including FFT and DFT parts, in hardware architecture. DFT core is used to deal with FFT points with the mixed powers of 2, 3, and 5 whereas FFT core is used to deal with FFT points with power of 2 only. Without making a good use of hardware reuse (DFT and FFT cores), it causes hardware waste and high area costs. Reference [23] uses radix-5 SDF-FFT basis to provide 46 FFT operating modes. Unfortunately, [13], [22], and [23] only cover up to $N = 2048$ FFT points. They have no further chances to apply for FFT points with $N > 2048$ if high FFT analysis resolution is

included in high-quality performance demands. Instead, our work can support up to 4096 FFT points (longest FFT length) and 48 FFT operating modes (the most number of FFT sizes) according to radix- 5^2 SDF-FFT basis.

In the core reconfigurable processing elements, other 7 design works only provide 2 - 7 different radix types of FFT operations. Correspondingly, the hardware flexibility is limited if the communication standard specification modifies or updates someday. Our work can provide up to 18 different radix types, which can fully reuse the hardware resource of complex adders and complex multipliers. Therefore, we can enhance the freedom of core processing hardware usage, especially for more radix types with the mixed powers of 2, 3, and 5 [see TABLE II again]. In the storage elements (FIFO) aspect, our work is superior to the approaches in [22] and [23]. It is because our proposed FIFO approach is well-designed and can be easily extended to any larger FFT length, such as $N > 2048$. [22] cannot handle N FFT points with the decomposed factor of 5. Reference [23] is only workable for $N \leq 2048$ when the FIFO sub-bank partition and distribution plan are only based on supporting $N = 2048$ maximally. Besides, our work can operate at the highest working frequency (such as 526.32 MHz) among all of 8 design works. There are two reasons. First, we make large efforts on modifying the processing-element circuits according to the original radix- 5^2 basis and developing the hardware-reduced version (CR-PE). Second, by carefully analyzing the hardware component connections and suitably inserting pipelining stages, the purpose of the critical-path reducing, i.e., speed enhancing, is successfully achieved [also see Fig. 3 again].

From different design approaches and technology applied, it's better to make comparisons with existing performance indexes, such as NAE and NEE [32]. It is important that NAE and NEE do not taken the max FFT-length effects

TABLE XIII
CHIP COMPARISON WITH OTHER DESIGN WORKS IN CURRENT LITERATURE (FOR 3GPP-LTE APPLICATIONS)

Design Works	Proposed Work	ISCAS'16 [12]	TVLSI'13 [11]	TVLSI'15 [19]	JSSC'12 [24]	TCAS-I'17 [22]	TVLSI'17 [13]	TCAS-I'18 [23]
FFT Sizes Supported in 3GPP LTE std.	4 - 4096	16/64/256/1024	128/512/1024/2048	128/256/512/1024/1536/2048	128/256/512/1024/1536/2048	4 - 2048 *	12 - 2048	4 - 2048
Number of Modes	48	4	4	6	6	32 *	40	46
Architecture Base	SDF	SDC/SDF	MDC	SDF	SDF	SDF	Memory-Based	SDF
Number of Radix Types	18	2 (R4, 8)	2 (R4, 8)	4 (R2, 3, 4, 8)	3 (R2, 6, 8)	6	4 (R2, 3, 4, 5)	7
Process Technology	40 nm	65 nm	90 nm	90 nm	65 nm	90 nm	55 nm	40nm
Supply Voltage (V)	0.99	1.0	1.0	0.9	0.45	1.0	1.08	0.99
Max. FFT Length	4096	1024	2048	2048	2048	2048 *	2048	2048
Wordlength (Bits)	14	32	10	12	12	16	16	14
Frequency (MHz)	526.32	400.00	40.00	40.00	20.00	188.67	122.88	500.00
Core Area (mm ²)	0.414	1.190	3.100	0.783	1.375	1.664	1.678	0.360
Gate Count	445 K	747 K	(NA)	205 K	1100 K	396 K	476 K	380K
Power Dissipation (mW)	63.9 @ N=1536 69.0 @ N=2048 73.3 @ N=3072 78.1 @ N=4096	35.7 - 416.8	51.69 @ N=128 57.51 @ N=512 62.92 @ N=1024 63.72 @ N=2048	2.35 @ N=128 2.64 @ N=256 3.08 @ N=512 5.48 @ N=1024 5.04 @ N=1536 6.43 @ N=2048	8.55	35.20	26.3 @ N=1296 19.2 @ N=2048	48.46
Throughput (M Symbols/s)	526.32	400.00	78.10	(NA)	20.00	188.67	(NA)	500.00
NAE (bits/mm ²) ^[32]	0.23	0.21	0.30	(NA)	0.18	0.29	(NA)	0.26
NEE (nJ/bit) ^[32]	2.30	6.00	2.45	(NA)	12.16	0.56	(NA)	1.50
Normalized Speed-Area Ratio (NSAR)	4844.5	548.3	(NA)	(NA)	37.2	975.7	(NA)	2694.7
Normalized Power-Frequency Ratio (NPFRR)	36×10^{-6}	1018×10^{-6}	778×10^{-6}	78×10^{-6}	209×10^{-6}	91×10^{-6}	105×10^{-6}	47×10^{-6}

Note : * This work is not dedicated to 3GPP-LTE applications. The original design can provide 48 modes (2-2187 points) in total, not limited to the FFT sizes in 3GPP-LTE applications. But it only supports 32 modes (4-2048 points) pre-defined in 3GPP-LTE applications.

into consideration. We just make a simple design reference and discussion here. By using NAE (area performance index in [32]), our chip is a little worse than [11], [22], and [23]. The design in [11] only performs 4 FFT modes and the according FFT length is limited to the power of 2. Reference [22] only supports up to 2187 FFT points and performs the FFT points with mixed powers of 2 and 3. In the processing kernel engine in [22], the hardware resource of complex adders and multipliers has less circuit area than ours. The reason is that supporting radix-2^X, radix-3^Y, and radix-5^Z is more complicated than supporting radix-2^X and radix-3^Y in nature. In nature, [23] requires less hardware resource of storage elements/bits while only providing up to $N = 2048$. On the other hand, by using NEE (energy performance index in [32]), our chip is only a little worse than [22] and [23] because our

work supports longer FFT length (such as twice of FFT length in both [22] and [23]).

In order to consider the influence with max FFT-length supporting, we make the according design comparison with other point-of-views by creating new performance indexes/ indicators, such as NSAR and NPFRR in (3) and (4), respectively. The motivation of removing the FFT-length effect is that there are more storage elements/bits on demand in hardware design for longer FFT-length supporting. As defined in (3), NSAR, normalized speed-area ratio, denotes gate efficiency for each FFT-point processing. If NSAR gets larger, it indicates that the circuit gates for every FFT point can deliver more system throughput. On the other hand, normalized power-frequency ratio (NPFRR) indicates green energy for each FFT-point processing as shown in (4). If NPFRR

becomes smaller, it denotes that the power consumption for every FFT point is lower at the same working frequency. They are used to make the normalization with respect to max FFT-length supporting, resulting in relatively fair comparison outcomes. By both two performance indexes/indicators, our work has the best system performance among all of 8 design works

$$\begin{aligned} & \text{Normalized speed – area ratio} \\ &= \frac{\text{Throughput}}{\left(\frac{\text{Gate Count}}{\text{MaxFFT Length}}\right)} \end{aligned} \quad (3)$$

$$\begin{aligned} & \text{Normalized power – frequency ratio} \\ &= \frac{\text{Power}}{(\text{Freq.} * \text{Max FFT Length})} \end{aligned} \quad (4)$$

V. CONCLUSION

We propose a flexible and reconfigurable FFT hardware architecture, of which the FFT kernel engine can perform 18 different changeable radixes. As compared to other existing FFT design works, our design work can perform larger FFT length (up to 4096 FFT points) and more FFT modes (48 different FFT sizes). The benefit is that covering higher FFT points can fulfill the applications with higher quality demands. Three developed design techniques are also useful for multi-mode and reconfigurable FFT system designs. In hardware implementation, an FFT ASIC realized with TSMC 40-nm CMOS technology only occupies a core area of 0.414 mm² and dissipates average power of 49.8 mW while working frequency operates at a maximum of 526.32 MHz. Moreover, our proposed design is competitive in terms of area cost and power/energy performance indexes, providing powerful results for the practical use of 3GPP-LTE communication system.

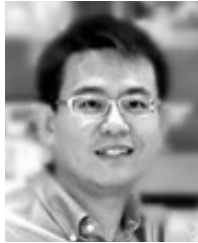
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